

MV SWGR 2190 DAYS/TA PM CHECKLIST

EQ#: _____ Unit: _____ SN# _____ WO# _____ Date: _____ PM Cycle: 2190 Days or each T/A

EQ	PM Item	Standard	Result	Comment		
Main Unit & LV Compartment	1 Appearance	Check for abnormal noise, dust, water, smoke, corrosion, cracks or inflammable vapors	PASS ☐ Abnormal ☐			
		Check gauges/meter	PASS ☐ Abnormal ☐			
		Check switches	PASS ☐ Abnormal ☐			
		Check indication lights	PASS ☐ Abnormal ☐			
		Check all label (Relay, Name tag, Arc flash, Testing/Calibration date and warning sign label)	PASS ☐ Abnormal ☐			
	2 wiring and Conduit	Check the wire and conduit entry seals	PASS ☐ Abnormal ☐			
Surge Arrestor	3 Cleaning	Clean the stationary portion by wiping with a clean cloth. Clean insulation with a dry cloth, dry vacuum, or OSHA approved solvent	PASS ☐ Abnormal ☐			
	1 Appearance	Check for abnormal noise, dust, water, smoke, corrosion, cracks or inflammable vapors	PASS ☐ Abnormal ☐			
	2 Cable and Conduit	Check the cable and conduit entry seals	PASS ☐ Abnormal ☐			
	3 Insulation Test	13.8KV: 800MΩ (5KV test 60s)	Phase-Phase (M Ω):			
			<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 33%;">A-B</td> <td style="width: 33%;">B-C</td> <td style="width: 33%;">C-A</td> </tr> <tr> <td style="height: 20px;"></td> <td></td> <td></td> </tr> </table>		A-B	B-C
A-B	B-C	C-A				
Phase-Ground (M Ω):						
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A-G	B-G	C-G				
4 High-Potential Test/ (PF test)	4.16KV: 14KV test voltage / 13.8KV: 27KV test voltage	PASS ☐ Abnormal ☐				
5 New replacement	Every 15 years	No replacement needed ☐ Replaced ☐				
	Installation date					
	Years in service					

MV SWGR 2190 DAYS/TA PM CHECKLIST - Breaker

EQ	PM Item	Standard	Result	Comment																					
Breaker	1	Trip coil bearing	Disassemble and clean or replace	PASS ☐ Abnormal ☐																					
	2	Arm Bushing	Check for damage, dust, discoloration or corrosion, engagement depth as per the vendor's standards	PASS ☐ Abnormal ☐																					
	3	Automatic Shutter	Check for damage, dust, discoloration or corrosion, shutter should work smoothly	PASS ☐ Abnormal ☐																					
	4	Breaker Rail	Check for damage, dust, or corrosion	PASS ☐ Abnormal ☐																					
	5	Insulation Resistance Test	13.8kV CB Test @ 5kV 60s, resistance $\geq 1000 \text{ M}\Omega$	<table border="1" style="display: inline-table; border-collapse: collapse;"> <tr> <td rowspan="2" style="padding: 5px;">Open</td> <td style="padding: 5px;">A - A</td> <td style="padding: 5px;">B - B</td> <td style="padding: 5px;">C - C</td> </tr> <tr> <td style="height: 20px;"></td> <td></td> <td></td> </tr> <tr> <td rowspan="3" style="padding: 5px;">Close</td> <td style="padding: 5px;">A - B</td> <td style="padding: 5px;">A - C</td> <td style="padding: 5px;">B - C</td> </tr> <tr> <td style="height: 20px;"></td> <td></td> <td></td> </tr> <tr> <td style="padding: 5px;">A - G</td> <td style="padding: 5px;">B - G</td> <td style="padding: 5px;">C - G</td> </tr> <tr> <td style="height: 20px;"></td> <td></td> <td></td> </tr> </table>	Open	A - A	B - B	C - C				Close	A - B	A - C	B - C				A - G	B - G	C - G				
	Open	A - A	B - B	C - C																					
	Close	A - B	A - C	B - C																					
A - G		B - G	C - G																						
6	Timing Test	VCB: Open $\leq 45 \text{ ms}$, Close $\leq 70 \text{ ms}$ VCS: Open $\leq 70 \text{ ms}$, Close $\leq 170 \text{ ms}$	PASS ☐ Abnormal ☐ <table border="1" style="display: inline-table; border-collapse: collapse;"> <tr> <td style="padding: 5px;">Open Speed (ms)</td> <td style="width: 100px; height: 20px;"></td> </tr> <tr> <td style="padding: 5px;">Close Speed (ms)</td> <td style="width: 100px; height: 20px;"></td> </tr> </table>	Open Speed (ms)		Close Speed (ms)																			
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7	Vacuum bottle Test	Pressure $\geq 1.000 \times 10^{-2} \text{ Pa}$	PASS ☐ Abnormal ☐ <table border="1" style="display: inline-table; border-collapse: collapse;"> <tr> <td style="padding: 5px;">A Phase</td> <td style="padding: 5px;">B Phase</td> <td style="padding: 5px;">C Phase</td> </tr> <tr> <td style="height: 20px;"></td> <td></td> <td></td> </tr> </table>	A Phase	B Phase	C Phase																			
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8	Contact resistance test	Low resistance test, resistance $< 100 \mu\Omega$	PASS ☐ Abnormal ☐ <table border="1" style="display: inline-table; border-collapse: collapse;"> <tr> <td style="padding: 5px;">A Phase</td> <td style="padding: 5px;">B Phase</td> <td style="padding: 5px;">C Phase</td> </tr> <tr> <td style="height: 20px;"></td> <td></td> <td></td> </tr> </table>	A Phase	B Phase	C Phase																			
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Action:

MV SWGR 2190 DAYS/TA PM CHECKLIST – CABLE

EQ	PM Item	Standard	Result	Comment
Cable	1 Cable testing (13.8KV)	Insulation test (5KV megger testing, to ground) Good: $\Omega > 800 \text{ M}\Omega$ Acceptable: $\Omega > 400 \text{ M}\Omega$ Tracking: $\Omega > 200 \text{ M}\Omega$	PASS ☐ Tracking ☐ Abnormal ☐ A-G: _____ Ω B-G: _____ Ω C-G: _____ Ω	
		Megger phase to phase no short	PASS ☐ Abnormal ☐	
		VLF insulation test (VLF-TD testing)	PASS ☐ Abnormal ☐	
		VLF insulation test report (Give HRE file to engineer, engineer generate report)	Report done ☐	
	2 Ground Test	Ground resistance $< 5 \Omega$	PASS ☐ Abnormal ☐ G: _____ Ω	
	3 Cable appearance	Check for damage, abnormal noise, dust, water, smoke, corrosion, cracks, discoloration, brittle insulation, loose shield ground, or presence of inflammable vapors.	PASS ☐ Abnormal ☐	
		Cable Terminal Block Check for loose connections	PASS ☐ Abnormal ☐	

Action:

MV SWGR 2190 DAYS/TA PM CHECKLIST – Relay

EQ	PM Item	Standard	Result	Comment	
Relay	1	Appearance	No abnormal noise, dust, smoke, corrosive, crack or inflammable vapors	PASS ☐ Abnormal ☐	
	2	Check Setting	Parameter should follow with the drawing	PASS ☐ Abnormal ☐	
	3	Trip protection Test	Send the trip signal to confirm that the breaker operates properly.	PASS ☐ Abnormal ☐	
Action:					

Rev. 0

Manager: _____ **Engineer :** _____ **Supervisor:** _____ **Inspect By:** _____

MV SWGR 2190 DAYS/TA PM CHECKLIST

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EQ	PM Item	Standard	Result	Comment											
PT ☐ in server	1 Appearance	Check for abnormal noise, dust, water, smoke, corrosion, cracks	PASS ☐ Abnormal ☐ N/A ☐												
	2 Cable and Conduit	Check the cable and conduit entry seals	PASS ☐ Abnormal ☐ N/A ☐												
	3 PT drawer finger	Check the PT finger spring tension and the contact tightness Note: Only applies to drawer-type PT	PASS ☐ Abnormal ☐ N/A ☐												
	4 Insulation Test	4.16KV: 400MΩ (2.5KV test 60s) 13.8KV: 800MΩ (5KV test 60s) Secondary to ground: 50 MΩ or higher (500 V test 60s) Note: This item shall be carried out when the power fuse blows.	Primary to ground (MΩ): <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th style="width: 33%;">A-G</th> <th style="width: 33%;">B-G</th> <th style="width: 33%;">C-G</th> </tr> <tr> <td style="height: 30px;"></td> <td></td> <td></td> </tr> </table> Primary to secondary (MΩ): <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th style="width: 33%;">A-A</th> <th style="width: 33%;">B-B</th> <th style="width: 33%;">C-C</th> </tr> <tr> <td style="height: 30px;"></td> <td></td> <td></td> </tr> </table>	A-G	B-G	C-G				A-A	B-B	C-C			
A-G	B-G	C-G													
A-A	B-B	C-C													
5 Tan δ Test	Below 5 % (Under 20°C) Note: Perform this inspection in Year 10, Year 13, and annually after the replacement interval	No replacement needed ☐ Replaced ☐													
		PT Installation date													
		PT Years in service													
CT ☐ in server	1 Appearance	Check for abnormal noise, dust, water, smoke, corrosion, cracks	PASS ☐ Abnormal ☐ N/A ☐												
	2 Cable and Conduit	Check the cable and conduit entry seals	PASS ☐ Abnormal ☐ N/A ☐												
	3 CT Test	1. 4.16KV: 400MΩ (2.5KV test 60s) 2. 13.8KV: 800MΩ (5KV test 60s)	Phase-Phase (MΩ): <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th style="width: 33%;">A-B</th> <th style="width: 33%;">B-C</th> <th style="width: 33%;">C-A</th> </tr> <tr> <td style="height: 30px;"></td> <td></td> <td></td> </tr> </table> Phase-Ground (MΩ): <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th style="width: 33%;">A-G</th> <th style="width: 33%;">B-G</th> <th style="width: 33%;">C-G</th> </tr> <tr> <td style="height: 30px;"></td> <td></td> <td></td> </tr> </table>	A-B	B-C	C-A				A-G	B-G	C-G			
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Bus	1	CB Secondary Bus Insulation Test	1. 4.16KV: 400MΩ (2.5KV test 60s) 2. 13.8KV: 800MΩ (5KV test 60s) Note: Phase-to-phase testing requires the cables to be disconnected. If they cannot be disconnected, perform phase-to-ground testing only.	Phase-Phase (MΩ): <table border="1"> <tr> <td>A-B</td> <td>B-C</td> <td>C-A</td> </tr> <tr> <td></td> <td></td> <td></td> </tr> </table> Phase-Ground (MΩ): <table border="1"> <tr> <td>A-G</td> <td>B-G</td> <td>C-G</td> </tr> <tr> <td></td> <td></td> <td></td> </tr> </table>	A-B	B-C	C-A				A-G	B-G	C-G				
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2	Main Bus	Contact resistance test, DC 100A, three phase resistance should be between 20% (±20%) Note: Main bus needs to be de-energized. If it cannot be de-energized, please enter N/A.	PASS ☐ Abnormal ☐ N/A ☒														
3	Busbar Connection Between SWGR Section	1. Torque all the bolts to correct spec Note: Main bus needs to be de-energized. If it cannot be de-energized, please enter N/A.	PASS ☐ Abnormal ☐ N/A ☐														
		2. Contact resistance test (DC 100A, three phase resistance should be between 20%) Note: Main bus needs to be de-energized. If it cannot be de-energized, please enter N/A.	PASS ☐ Abnormal ☐ N/A ☐														

ACTION:

Manager: _____ **Engineer :** _____ **Supervisor:** _____ **Inspect By:** _____